

dbGaP Data Access Request

Comprehensive Single-Cell Atlas for Non-Negative Matrix
Factorization-Based Transcriptomic Reference Construction

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1. Project Description

1.1. Background and Significance

Single-cell and single-nucleus RNA sequencing (scRNA-seq / snRNA-seq) has transformed our understanding of cellular heterogeneity in health and disease. Publicly deposited datasets in NCBI GEO now exceed 180,000 samples, yet a substantial fraction of the most clinically valuable single-cell data—particularly from studies involving human subjects with protected health information—resides in controlled-access repositories under dbGaP.

We have developed a scalable pipeline (**scgeo**) that discovers, reprocesses, and harmonizes single-cell FASTQ data into splicing-resolved count matrices using a uniform reference and quantification workflow (STARsolo / Cell Ranger / Salmon alevin-fry). The harmonized count matrices are then decomposed via **Non-Negative Matrix Factorization (NMF)** into a genes-by-factors matrix **W** and a factors-by-cells matrix **H**, which together define a universal transcriptomic reference that captures conserved gene programs across tissues, diseases, and developmental contexts.

Incorporating the controlled-access data in this reference is critical because:

- Disease-specific cohorts (cancer, neurodegeneration, autoimmune) are disproportionately represented in dbGaP relative to GEO.
- Paired whole-genome sequencing (WGS) and single-cell transcriptome data—available in studies such as HTAN, GTEx, and BICCN—enable linking genomic variation to transcriptomic cell states, a key requirement for integration with emerging foundation models (e.g., Google AlphaGenome).
- Atlas-scale programs (HuBMAP, KPMP, SenNet) provide comprehensive healthy-tissue references that anchor disease-state comparisons.

1.2. Research Use Statement

The requested individual-level data will be used exclusively for the following non-commercial, academic research purpose:

To construct a comprehensive, splicing-resolved single-cell transcriptomic reference by reprocessing raw FASTQ files from controlled-access single-cell and single-nucleus sequencing studies into harmonized count matrices, and decomposing those matrices via

Non-Negative Matrix Factorization (NMF) to identify universal gene expression programs. The resulting factor model ($\mathbf{W} \in R^{G \times K}$, $\mathbf{H} \in R^{K \times N}$) captures population-level gene programs without retaining individual-level identifiable information. Only the $\mathbf{W}$ matrix (gene loadings) and aggregated per-factor summary statistics will be distributed; individual-level $\mathbf{H}$ coefficients, raw reads, and donor metadata will remain within the approved secure compute environment.

1.3. Specific Aims

1. **Aim 1: Harmonized Reprocessing.** Download raw FASTQ files for all requested studies to the approved university HPC cluster. Align reads to a uniform reference genome (GRCh38 / GRCm39) using STARsolo with splice-aware quantification, producing per-sample count matrices with spliced / unspliced layers.
2. **Aim 2: NMF Decomposition.** Integrate the reprocessed count matrices with our existing public-data catalog (~180K samples from GEO + 894 experiments from ENCODE) and perform rank-optimized NMF to derive a universal gene-program dictionary $\mathbf{W}$.
3. **Aim 3: Genome–Transcriptome Linkage.** For studies with paired WGS/WES and single-cell transcriptome data (e.g., HTAN, GTEx, BICCN), project individual cells onto the NMF factor space and associate factor usage patterns with germline and somatic variants to enable integration with genomic foundation models.
4. **Aim 4: Public Reference Release.** Release only the non-identifiable $\mathbf{W}$ matrix and aggregated factor-level summary statistics (k-anonymized, $n \geq 10$ donors per group) as a public transcriptomic reference. No individual-level data, $\mathbf{H}$ matrix values, or donor metadata will be redistributed.

2. Requested Datasets

We request access to the following 121 dbGaP studies, grouped by consortium and disease focus. All studies contain single-cell or single-nucleus sequencing data with raw FASTQ files deposited in SRA under controlled access.

2.1. Major Consortium Studies

phs Accession	Consortium	Description
phs002371	HTAN	Human Tumor Atlas Network — pan-cancer SC + spatial + WGS
phs000424	GTEx	Genotype-Tissue Expression — snRNA-seq + WGS across 8 tissues
phs002673	BICCN	Brain Initiative Cell Census Network — 50M+ brain cells
phs003431	BICAN	Brain Initiative Cell Atlas Network — next-gen brain atlas
phs003604	BICAN-2	BICAN Phase 2
phs001810	HuBMAP	Human BioMolecular Atlas Program — SC across organs
phs002507	HuBMAP-2	HuBMAP Phase 2

phs Accession	Consortium	Description
phs000178	GDC/TCGA	The Cancer Genome Atlas — WGS/WES + SC
phs000218	TARGET	Therapeutically Applicable Research — pediatric cancers
phs001211	TOPMed	Trans-Omics for Precision Medicine — WGS + phenotypes
phs002362	HTAN-HTAPP	Human Tumor Atlas Pilot — SC + WGS
phs002624	SenNet	Cellular Senescence Network — SC atlas of aging
phs002256	KPMP	Kidney Precision Medicine Project — snRNA + spatial
phs003169	HCA	Human Cell Atlas — large-scale SC reference
phs002716	AMP-PD	AMP Parkinson’s Disease — WGS + snRNA
phs001515	LungMAP	NHLBI Lung Molecular Atlas Program
phs002062	4DN	4D Nucleome — chromatin architecture
phs003366	IGVF	Impact of Genomic Variation on Function
phs003123	SPARC	Stimulating Peripheral Activity
phs002159	HPAP	Human Pancreas Analysis Program
phs002279	ABI	Allen Brain Institute Cell Types
phs001246	SEA-AD	Seattle Alzheimer’s Disease Brain Cell Atlas

2.2. Individual Investigator Studies

The remaining 99 phs accessions represent individual investigator-led studies spanning cancer (36 studies), neurological disease (28), immune/autoimmune (15), developmental biology (10), and other categories (10). A complete list with phs accessions and titles is provided in **Supplementary Table S1** appended to this application.

2.3. Data Types Requested

For each study we request:

- **Raw sequencing data (FASTQ/BAM):** Required for uniform reprocessing with our harmonized pipeline.
- **Sample and donor metadata:** Cell type annotations, tissue of origin, disease status, age, sex, treatment—needed for meaningful NMF factor annotation. All metadata remains within the secure compute environment.
- **Genotype/WGS data (where available):** For the 153 studies with paired WGS/WES data, used exclusively for genome–transcriptome linkage analysis (Aim 3).

3. Data Security and Institutional Oversight

3.1. Institutional Affiliation

Institution: *[University Name]*
Signing Official: *[Name, Title]*
IT Security Officer: *[Name, Title]*
IRB of Record: *[IRB Protocol Number]*

The Principal Investigator has completed the following:

- NIH Genomic Data Sharing (GDS) Responsible Conduct of Research training
- CITI Human Subjects Research training
- Institutional Security Awareness training

3.2. IT Security Plan

Data will be stored and processed on the university's **High-Performance Computing (HPC)** cluster, which meets or exceeds the NIH Security Best Practices for Controlled-Access Data:

Requirement	Implementation
Physical security	University data center with badge access, 24/7 surveillance
Network security	Campus firewall, intrusion detection (IDS/IPS), VPN-only access
Authentication	University SSO with MFA; SSH key-based access to HPC
Authorization	POSIX ACLs restricting data directories to PI + approved personnel
Encryption at rest	AES-256 on GPFS/Lustre filesystem or encrypted scratch
Encryption in transit	TLS 1.3 for all data transfers; SCP/SFTP for file movement
Logging & auditing	All access logged via <code>auditd</code> ; quarterly review
Data destruction	Secure deletion (NIST 800-88) upon project completion or expiry
Incident response	University CISO incident response plan; 24-hour breach notification

3.3. Data Use Limitations

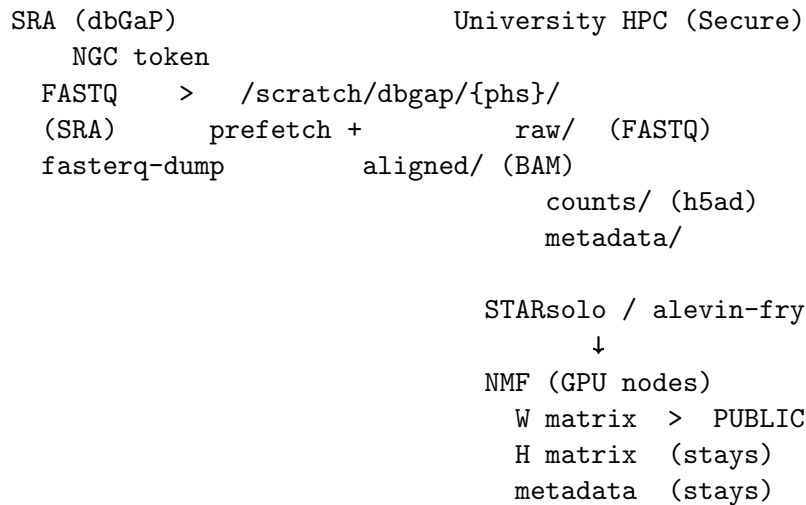
- **No redistribution** of individual-level data (FASTQ, BAM, count matrices, metadata).
- **No re-identification attempts** of study participants.
- **No cloud processing** — all data remains on university on-premises HPC infrastructure.
- **Outputs released publicly** are limited to the population-level **W** matrix and k-anonymized ($k \geq 10$) summary statistics.
- Data access will be terminated and data securely destroyed upon project completion, change of PI affiliation, or revocation of dbGaP approval.

4. Compute Architecture and Data Flow

4.1. Processing Pipeline

1. **Download:** `prefetch` + `fasterq-dump` from SRA to university HPC scratch storage via authenticated dbGaP download (NGC token).
2. **Alignment:** STARsolo (10x Chromium, Drop-seq, inDrop) or Salmon alevin-fry (Smart-seq2/3) on compute nodes. SLURM job scheduler, 64–128 cores per job.
3. **QC & Filtering:** Per-cell QC metrics (UMI counts, gene counts, mitochondrial fraction); doublet detection (Scrublet).
4. **Count matrix output:** Sparse matrices (genes $\times$ cells) with spliced/unspliced layers stored in `.h5ad` format on encrypted storage.
5. **NMF training:** GPU-accelerated rank-optimized NMF on the merged count matrix. The **W** matrix is exported; **H** remains on secure storage.
6. **Data destruction:** Raw FASTQ and intermediate files deleted upon successful QC of count matrices. Count matrices retained only for the duration of the approved project period.

4.2. Data Flow Diagram



5. PHI and Privacy Considerations

5.1. Output Privacy Analysis

The sole public output of this project is the $\mathbf{W} \in R^{G \times K}$ matrix, where $G \approx 33,000$ genes and $K \approx 100\text{--}500$ factors. This matrix represents **population-level gene program definitions** analogous to a dictionary of gene co-expression patterns. It contains:

- No individual-level data
- No donor identifiers or demographic information
- No cell-level expression values
- No sequence-level information (no reads, no variants)

The **H** matrix (factors $\times$ cells) and all donor metadata remain within the secure compute environment and are never distributed.

Summary statistics derived from $\mathbf{H}$ are released only after k-anonymization: for each NMF factor, we report mean and variance of factor usage across coarse-grained groups (tissue type, broad disease category) with a minimum group size of $k = 10$ donors. This satisfies standard k-anonymity requirements and prevents re-identification.

5.2. Re-identification Risk Assessment

The $\mathbf{W}$ matrix is a **generative model parameter** rather than a transformation of individual records. Formally, $\mathbf{W}$ is the solution to:

$$\min_{\mathbf{W} \geq 0, \mathbf{H} \geq 0} \|\mathbf{X} - \mathbf{WH}\|_F^2 + \lambda \|\mathbf{H}\|_1$$

where $\mathbf{X}$ is the full count matrix. Since $\mathbf{W}$ is determined by the global structure of $\mathbf{X}$ rather than any individual column, removing or adding a single donor changes $\mathbf{W}$ by $O(1/N)$ where N is on the order of millions of cells. Individual contribution is negligible and non-recoverable.

6. Personnel

Name	Role	Responsibilities	Training
Zach DeBruine, PhD	PI	Study design, NMF methods, oversight	GDS, CITI
[TBD]	Co-I	Pipeline development, QC	GDS, CITI
[TBD]	Data Analyst	Reprocessing, metadata	GDS, CITI

All personnel with data access will complete required training prior to data download. Access will be revoked immediately upon departure from the approved institution.

7. Timeline

Milestone	Target
DAR approval	T_0
Infrastructure validation & test downloads	$T_0 + 2$ weeks
Phase 1: Consortium studies (HTAN, GTEx, BICCN, HuBMAP)	$T_0 + 3$ months
Phase 2: Disease-specific studies (cancer, neuro)	$T_0 + 6$ months
Phase 3: Remaining studies + paired WGS analysis	$T_0 + 9$ months
NMF model training and validation	$T_0 + 12$ months
Data destruction (raw files)	$T_0 + 14$ months
Annual renewal / closeout	$T_0 + 12$ months

Supplementary Table S1: Complete phs Accession List

phs Accession	Study Title
phs000007	NHLBI Framingham SABRe CVD
phs000159	Therapeutic targets in acute myeloid leukemia

phs Accession	Study Title
phs000178	The Cancer Genome Atlas (GDC/TCGA)
phs000187	High Density SNP Association Analysis of Melanoma
phs000218	Therapeutically Applicable Research (TARGET)
phs000363	NHLBI Framingham SABRe CVD
phs000424	Genotype-Tissue Expression (GTEx)
phs000549	Clonal Evolution of Pre-Leukemic HSC
phs000640	Massively Multiplex Single-Cell Hi-C
phs000748	Therapeutic targets — multi-omic characterization
phs000815	Gene Expression and Regulatory Networks in Leukocytes
phs000833	Single Cell Analysis Program – Transcriptome (SCAP-T)
phs000834	SCAP-T (UCSD)
phs000835	SCAP-T (U. Penn)
phs000836	SCAP-T (USC)
phs000924	iPSCORE (iPSC Collection for Omic Research)
phs000933	Acral melanoma molecular profiling
phs000958	CSER-MedSeq
phs000989	Human Developmental Neurogenesis
phs000993	GTEx Pilot
phs001016	Fixed Single Cell Study (Radial Glia)
phs001024	NHGRI GREGoR
phs001027	SC genomics — Homo sapiens
phs001029	Full-Length scRNA-seq (HeLa S3)
phs001048	FUSION Tissue Biopsy Study
phs001068	FUSION Muscle Expression
phs001110	Kids First — Congenital Heart Disease
phs001158	Spatial transcriptomics of DRG (pain)
phs001181	SC DNA and RNA Sequencing of CLL
phs001187	Epigenomics of CD8 T cell Aging
phs001188	FUSION Islet Expression
phs001200	West Nile Virus Neutralizing Antibodies
phs001205	hESC-derived brain cell lineages
phs001211	TOPMed
phs001231	NeuroLINCS
phs001246	Seattle Alzheimer’s Disease (SEA-AD)
phs001255	Molecular Evolution of Cancer
phs001268	Combinatorial Indexing SC Genomes
phs001269	Massively Multiplex SC Hi-C
phs001276	hESC-derived interneuron development
phs001294	SC genomics study
phs001300	Single Nuclei Multiome – NABEC Cortex
phs001323	Immune biomarkers of checkpoint response
phs001362	Linked Read Sequencing – Gastric Cancer
phs001372	Integrated SC Analysis in CLL
phs001378	Lymphoma scRNA-Seq
phs001438	Chromatin Accessibility in Developing Brain
phs001457	AMP-RA (Rheumatoid Arthritis)
phs001515	NHLBI LungMAP
phs001519	Neoantigen Vaccine in GBM
phs001525	X-Linked Dystonia Parkinsonism
phs001529	scRNA-Seq of RA Synovial Tissue
phs001543	VAST — Vascular Smooth Muscle
phs001587	Cellular heterogeneity across lineage

phs Accession	Study Title
phs001615	Epigenetic Landscape of RA FLS
phs001627	SC DNA Sequencing of AML
phs001671	U6 snRNA / LINE-1 Retrotransposition
phs001680	T Cell States and Checkpoint Blockade in Melanoma
phs001703	Genetic Polymorphisms and Immune Gene Expression
phs001704	Enhancer Mapping in CLL
phs001750	IPF Cell Atlas
phs001763	All of Us Research Program
phs001773	Linked-Read Sequencing — Complex SVs
phs001790	NIMH Human MTG Cell Types
phs001810	HuBMAP
phs001836	SC Transcriptomics of Neocortical Development
phs001840	SC Analysis of Pancreatic Ductal Adenocarcinoma
phs001854	Pilocytic Astrocytoma scRNA-Seq
phs001861	Uveal Melanoma SC Evolution
phs001886	SC Analysis of Human Placenta
phs001907	BRAIN Initiative SCAP-T Phase 2
phs001934	Sexual Dimorphism in Immune Aging
phs001961	SN Multiomic Profiling of Host Response
phs002003	Fetal Gene Expression Atlas
phs002007	Innate T Cell Transcriptomics
phs002008	Orofacial Morphology Enhancers
phs002015	SCAP-T Phase 2 (Broad)
phs002048	SC Analysis of Systemic Lupus
phs002062	4D Nucleome
phs002071	Donor Pancreata Transcriptomic Signatures
phs002159	Human Pancreas Analysis Program (HPAP)
phs002204	Cardiac Cell Type Gene Regulatory Programs
phs002256	KPMP
phs002258	HEAL Initiative
phs002279	Allen Brain Institute Cell Types
phs002280	Multimodal Brain Analysis
phs002316	Trimodal SC Measurement (DOGMA-seq)
phs002361	Baseline Immune and Checkpoint Response
phs002362	HTAN-HTAPP
phs002371	HTAN
phs002386	H3Africa
phs002407	MMRd/MMRp Colorectal Cancer Atlas
phs002443	In-situ to Invasive Breast Cancer
phs002507	HuBMAP Phase 2
phs002624	SenNet
phs002655	SARS-CoV-2 Immune Perturbations in Infants
phs002673	BICCN
phs002716	AMP-PD
phs002790	NCI/IOTN Immunotherapy
phs002808	NCI Moonshot — Tumor Atlas (TNET)
phs002915	SC Multi-omic Velocity — Gene Regulation
phs002922	Baseline Immune and T Cell Kinetics
phs002926	Infant Serological Responses
phs003002	CD137 Agonism + Anti-PD1
phs003015	Donor/Recipient Immune Phenotypes Post-Transplant
phs003123	SPARC

phs Accession	Study Title
phs003169	Human Cell Atlas
phs003230	Neuroendocrine Prostate Cancer Evolution
phs003260	SN Transcriptomes — Opioid Overdose
phs003366	IGVF
phs003417	Multimodal Gene Expression at Cell Level
phs003418	SC Characterization of Microscopic Colitis
phs003427	SC Profiling of Premature Neonate Airways
phs003431	BICAN
phs003604	BICAN Phase 2